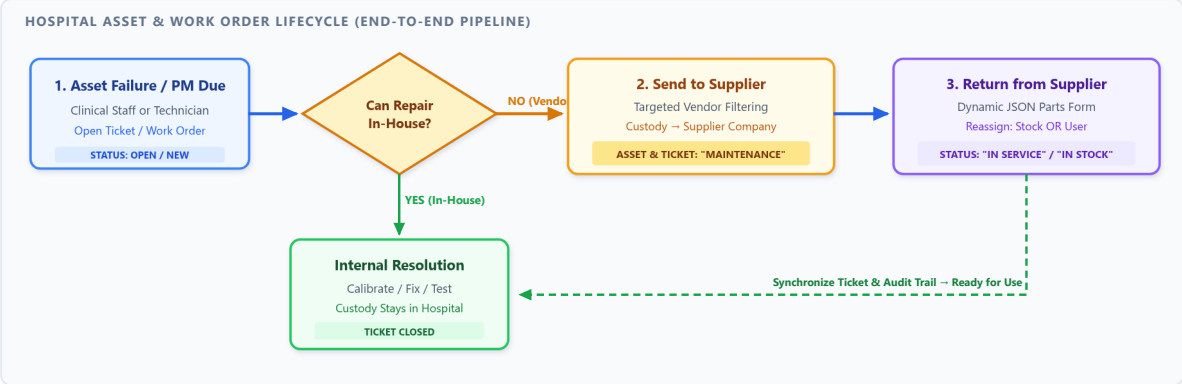
Hospital Equipment Maintenance System

End-to-End Operational Flowchart & Custody Lifecycle Architecture

Document: **SOP-ENG-001**
System: **Stock & Helpdesk Portal**
Theme: **Light Medical Theme**
Generated: **October 2026**

ARCHITECTURE 1. High-Level Maintenance Ecosystem Overview



RBAC POLICY 2. Role-Based Access Control (RBAC) & Team Specialization

Team / Role	Assigned Equipment Scopes	Maintenance & Work Order Privileges	Vendor Visibility Filter
ADMIN	Universal (All Hospital Assets)	Full CRUD, Work Orders, Custody Overrides, Global Reports	All Vendors (IT, BioMed, General)
MAINTENANCE_IT	Laptops, PCs, Servers, Switches, Printers	IT Work Orders, Specs Form Builder, Dispatch IT Hardware	Vendors with `"IT"` category
BIOMEDICAL	Ventilators, Monitors, Defibrillators, Pumps	Medical Device Calibrations, BioMed Repairs, Parts JSON	Vendors with `"BIOMEDICAL"` category
GENERAL	HVAC, Generators, Beds, Lighting, Facilities	Facility Work Orders, General Infrastructure Overhauls	Vendors with `"GENERAL"` category
END_USER (Clinical)	Custody-assigned equipment only	Open Ticket, Request Inspection, View Ticket Status (Read-Only)	None (Hidden from clinical staff)

1. Preventive (PM) SCHEDULED
Time-based inspection to prevent failure. Overhauls, calibration, filter replacements.

2. Corrective (CM) BREAKDOWN
Triggered by staff defect reports. Asset immediately flagged for triage and diagnosis.

3. In-House Triage INTERNAL
Repaired by hospital engineers within facility workshop. Custody remains internal.

4. Vendor Repair EXTERNAL
Sent out to manufacturer/supplier under official work order. Custody is legally transferred.

Phase 1: Dispatch to Supplier & Custody Handover

Hardware Triage, Qualified Vendor Filtering & Database State Synchronization

Route: **/Maintenance/SendToSupplier**
Status Transition: **In Service / In Stock → Under Maintenance**

DISPATCH WORKFLOW 1. Step-by-Step Equipment Dispatch Sequence

STEP 01 • IDENTIFICATION

Select Defective Asset

Technician opens asset view or defect ticket. Confirms device category, department, and serial number.

→

STEP 02 • PRE-SEND DIAGNOSIS

Fill Issue Description

Detailed defect symptom log is written (e.g., "Motherboard failure", "Calibration drift").

→

STEP 03 • SUPPLIER FILTERING

Choose Qualified Vendor

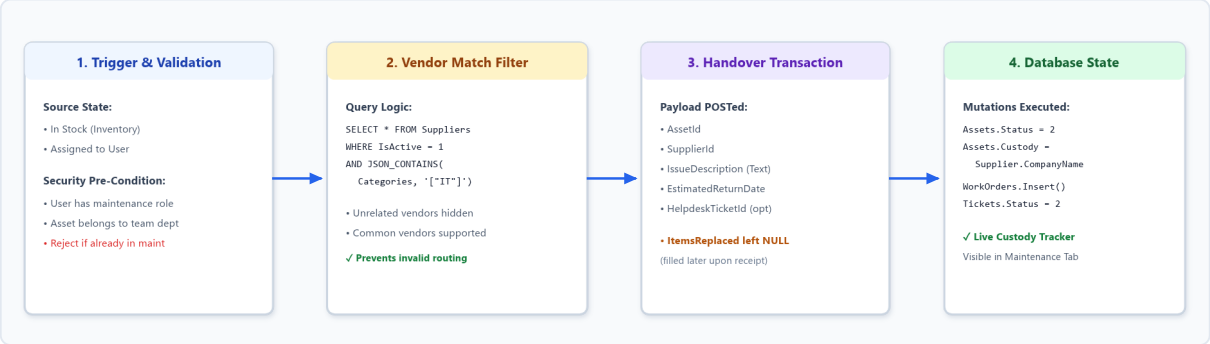
Dropdown auto-filters suppliers based on matching asset department (IT, BioMed, General).

→

STEP 04 • CUSTODY HANDOVER

Execute Work Order

Physical unit dispatched. Custody switches to Supplier Company. Ticket status → Maintenance.



FILTERING LOGIC 2. "Vendors That Can Help" Filtering Logic

Device Scope	Supplier Category	Example Vendors
IT Devices	`["IT"]`	Dell Enterprise, HP Service, Cisco Pro
BioMed Devices	`["BIOMEDICAL"]`	Siemens Healthineers, Philips Med, GE
Facilities	`["GENERAL"]`	Daikin HVAC, Schindler Elevators, Otis
Multi-Discipline	`["IT", "GENERAL"]`	Schneider Electric, Siemens Infra

AUDIT TRAIL 3. Database State Mutations on Dispatch (Phase 1)

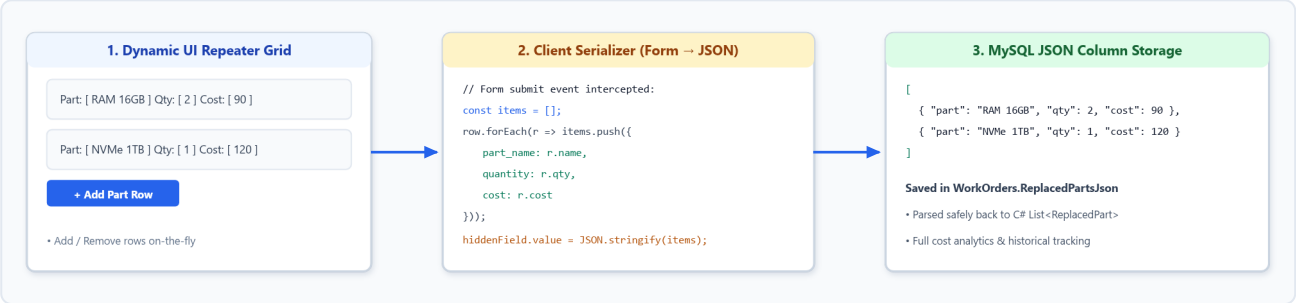
Table	Column	New Value
`Assets`	`Status`	`Maintenance` (Enum = 2)
`Assets`	`CurrentCustody`	`Supplier.CompanyName`
`MaintenanceOrders`	`Status`	`SentToSupplier`
`Tickets`	`Status`	`Maintenance`

Phase 3: Receiving, Replaced Parts JSON & Custody Reassignment

Hardware Check-in, Dynamic Form Builder, and Dual Custody Routing

Route: **/Maintenance/ReturnFromSupplier/{id}**
Status Transition: **Under Maintenance → In Service / In Stock**

JSON PIPELINE 1. Dynamic Replaced Parts JSON Builder Pipeline



ROUTING LOGIC 2. Custody Reassignment Decision Tree (Central Stock vs User)

When equipment is received from the supplier, custody MUST NOT stay with the vendor. The receiving officer chooses the target destination:

Option A: Return to Stock

- `Asset.CurrentCustody = "Stock"`
- `Asset.AssignedUserId = NULL`
- `Asset.Status = InStock`
- Ready for future redeployment

Option B: Return to User

- `Asset.CurrentCustody = User.FullName`
- `Asset.AssignedUserId = SelectedUserId`
- `Asset.Status = InService`
- Directly handed back to department

TICKET SYNC 3. Helpdesk Ticket Synchronization on Return

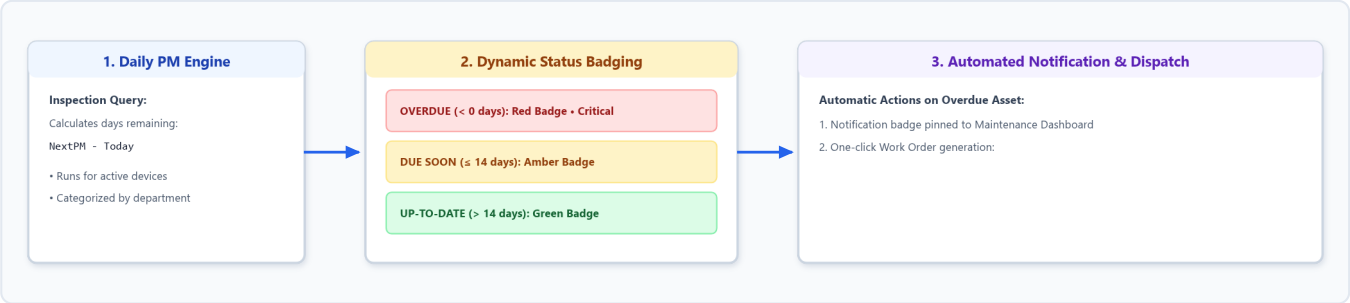
Entity	Attribute	Value on Return
`MaintenanceOrder`	`ActualReturnDate`	`DateTime.UtcNow`
`MaintenanceOrder`	`ItemsReplaced`	Structured JSON String
`MaintenanceOrder`	`Status`	`Completed`
`Ticket` (Linked)	`Status`	`Resolved` / `Closed`
`TicketComment`	`Body`	Auto-logged return note + part list

Preventive Maintenance (PM) & Technical Audit Trail

Inspection Intervals, Recurring Overhauls, and Historical Compliance

Module: **Preventive Maintenance Engine**
Compliance: **Medical Equipment Safety Overhaul**

INSPECTION ENGINE 1. Preventive Maintenance (PM) Interval Scheduling Flowchart



`/Maintenance/SendToSupplier` 3. Auto-computes Next PM Date on Return: `NextPM = Today + IntervalDays` ✓ Fully closes the recurring audit loop

ROUTING PATHWAYS 2. Routine In-House Checkup vs External Overhaul Pathway

Criteria	Pathway A: In-House	Pathway B: Supplier Overhaul
Scope	Filter swap, sensor zeroing, battery test	Internal motor rebuild, recalibration cert
Custody	Remains with Hospital / User	Transferred to Supplier Company
Status	Stays "In Service"	Switches to "Under Maintenance"
Parts Used	Hospital internal stock	Supplier items → ReplacedPartsJson
Certification	Signed by in-house engineer	Vendor certificate attached to Order

AUDIT TRAIL 3. Work Order Details View & Historical Compliance Log

Auditing & Safety Standard Compliance

COMPLIANT

Every maintenance operation creates an immutable historical event accessible via `/Maintenance/Details/{id}` and `/Assets/Details/{id}`:

- **Pre-Send Snapshot:** Date dispatched, technician user, defect log, targeted supplier.
- **Custody Chain:** Timestamped record of who possessed device at any historical date.
- **Replaced Parts Breakdown:** Granular inventory cost, part serials, and quantities.
- **Final Disposition:** Destination (Stock vs Clinical User) and ticket closure link.